ATS User Manual

Complete guide for installation, configuration, and operation of the Echo ATS Acoustic Test System

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Echo warrants its products to be free of manufacturing defects for a period of three years from the date of purchase. Any product covered under a valid Echo new product warranty—where the damage is not caused by owner misuse or abuse (see Warranty Exclusions, below)—will be repaired or replaced by Echo free of charge. **Please see the full Echo Warranty Policy for more information.**

Regulatory Compliance Information

Echo Digital Audio has conformed and this product conforms, where applicable, to the European Union's Directive 2002/95/EC on Restrictions of Hazardous Substances (RoHS) as well as the following sections of California law which refer to RoHS, namely sections 25214.10, 25214.10.2, and 58012, Health and Safety Code; Section 42475.2, Public Resources Code.

As with the disposal of all old electrical and electronic equipment, this product is not to be treated as regular household waste. Instead, it shall be handed over to the applicable collection point for the recycling of electrical and electronic equipment.

CE Compliant **RoHS** **WEEE**

Important Safety Instructions

> ***WARNING***

1. **Read Instructions** — Read all safety and operating instructions before operating this product
2. **Follow Warnings** — Observe all warnings on your Echo ATS and in this manual
3. **Moisture Protection** — Do not install or operate near water sources (sinks, damp basements, etc.)
4. **Heat Protection** — Keep away from heat sources such as heaters or radiators
5. **Power Requirements** — Operate only from stable AC power as specified. Use surge protector in areas prone to lightning or noisy factory environments
6. **Proper Grounding** — Use only the provided grounded plug. Do not modify the plug. Ensure outlet is properly grounded
7. **Power Cord Safety** — Route power cords to avoid damage to plugs, outlets, and cord exit points
8. **No User Servicing** — There are no user serviceable parts inside. Opening the case voids warranty and exposes you to hazardous voltages

Contact your authorized repair center for all servicing needs.

1. Introduction

The Echo ATS Test System is a USB 2.0 compliant audio test platform compatible with ASIO and WASPI drivers on Windows computers, Core Audio on Macintosh (Mac) computers, ALSA on Linux computers. When coupled with a computer running test software, the ATS forms a complete system for testing digital and analog audio electronics and acoustics in consumer and professional products. Echo ATS products are ideally suited for a high-volume production environment where testing occurs on a continual basis.

Echo ATS combines the functionality of multiple pieces of test equipment into a single, integrated device for increased reliability at a reduced cost. In addition to audio testing, it has the capability to control measurement fixtures through its GPIO interface.

Figure 1: Echo ATS Test System

2. Installation

Hardware Setup

Step 1: Connect Power

Connect the power cable to the IEC C14 power inlet. Both locking and non-locking cables may be used. The LED on the front panel is green when power is applied and the ATS is operating properly.

> ***Note:*** *The universal AC input accepts 100 – 240 VAC at 50 or 60 Hz.* ***The ATS, equipment connected to the ATS, and the test computer must all be connected to the same surge-protected power strip.***

Step 2: Connect USB

Connect the supplied USB-C cable to the USB-C connector on the back of the ATS, and the other end to the host computer.

> ***Tip - Mounting:*** *The ATS includes mounting holes on the bottom of the unit for secure installation. See the mounting holes drawing for dimensions and hole positions.*

Driver Installation

Windows Driver Installation

Download the ATS Windows driver and run the installer.

The installer includes:

- ATS Control Panel application
- Latest firmware updates
- ASIO driver (appears as "ASIO Echo ATS")
- Command-line utilities

> ***Note:*** *Although Windows 10/11 will automatically recognize the ATS using the built-in WASAPI driver, we strongly recommend installing the Echo software for:*
>
> - *Control panel functionality*
> - *Firmware update capability*
> - *ASIO driver support (required by most professional audio applications)*

macOS Setup

Support for USB audio is built into macOS through Core Audio. The ATS is automatically detected whenever it is powered on and connected to a Mac USB port.

No driver installation required — The ATS will appear immediately in your audio applications.

> ***Note:*** *While no driver is needed, you may want to download the macOS command-line utilities for programmatic control from Echo: ATS Downloads*

3rd Party Test and Measurement Software

The Echo ATS Test System may be used with a wide variety of 3rd party test and measurement software via the Echo ATS ASIO driver. Because ASIO is a digital interface, all levels are communicated between the hardware and software as digital values, and there's no inherent conversion to actual analog I/O values when using an analog interface. To properly do the conversion, the software needs to know the correct scaling factor, which defines the relationship between analog voltage and digital full-scale level (0 dBFS). The scaling factors for different types of inputs and outputs are typically different. For example, the scaling factor of a line-level output will be different than for a speaker output, because the maximum voltage of a speaker output is greater.

Although scaling factors can be determined by experimentation using a calibrated voltage meter, we provide you with the correct scaling factors for our hardware. You can simply enter these values into the appropriate settings in the measurement software that you are using.

Output Scaling Factors

IO Type	Scaling Factor
Line (unbalanced)	4.0 V/FS
Line (balanced)	8.0 V/FS
Headphone	3.0 V/FS
Amplifier/Speaker	9.545 V/FS

Input Scaling Factors

IO Type	Scaling Factor
Mic-Line	161.6 mFS/V
Line Impedance VMON	333.3 mFS/V
Line Impedance IMON *	8.0 FS/V
Speaker Impedance VMON	80.80 mFS/V
Speaker Impedance IMON *	161.6 mFS/V

* IMON measures current through the 0.1Ω sense resistor

3. ATS Control Panel

Overview

The control panel controls and monitors the following ATS hardware functions:

- Analog input and output gain
- Constant-current power to microphones
- TEDS data
- Level meters
- Firmware update
- WASAPI and ASIO driver settings
- Clock source and status
- Diagnostic logging and directory

The tabs and drop-down menus are described in the text that follows. If the control panel is not already installed, please see the *Echo* Driver Software section. **The control panel is for ATS hardware control and metering only. It does not perform any audio analysis.**

Channels Tab

The Echo Acoustic Test System Inputs and Outputs are configurable for testing different types of devices (i.e., acoustic, line level, headphone, or speaker). When you configure ATS I/O for testing different types of devices, it adjusts the relative voltage and phase based on the desired setup.

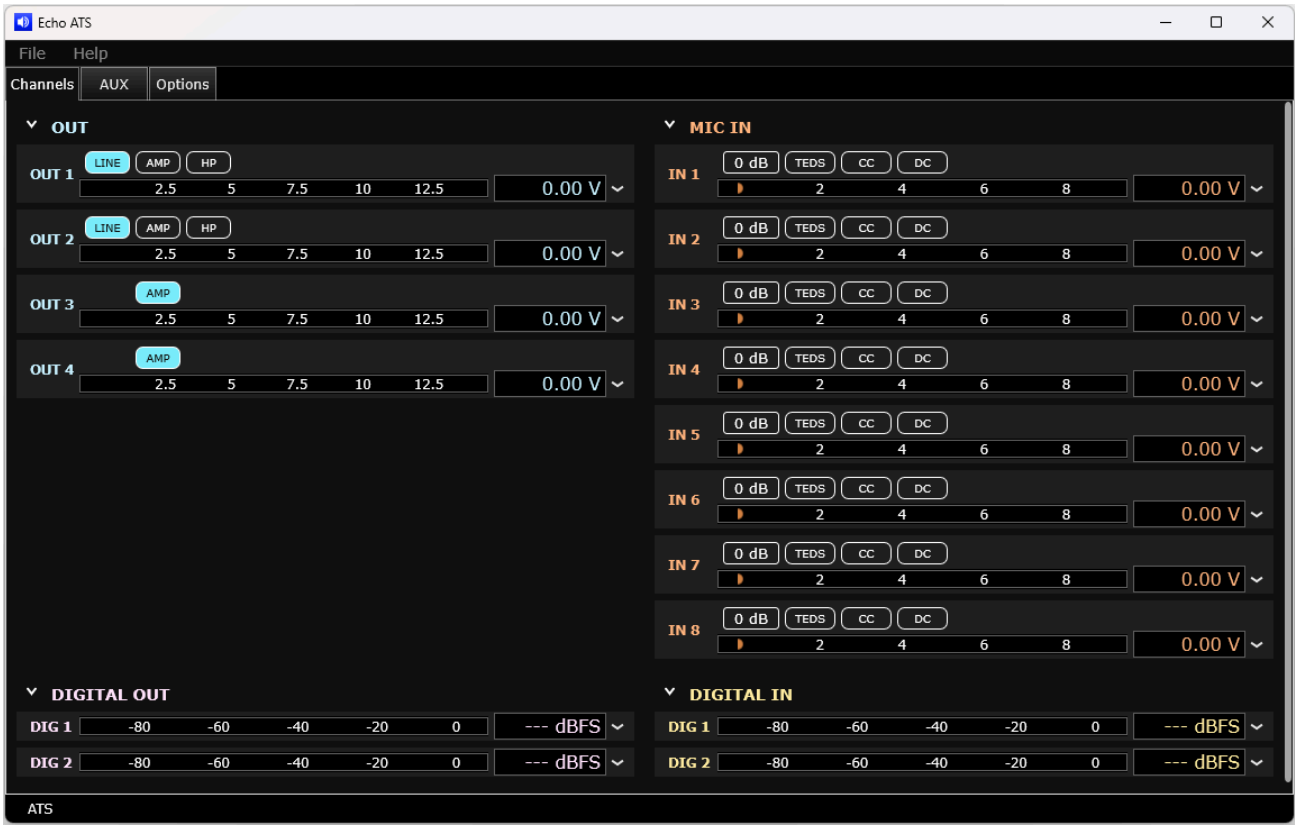

Figure 2: Channels tab

4. Analog Outputs

Overview

The ATS provides four configurable analog output channels with three distinct operating modes optimized for different test scenarios. Channels 1 and 2 are fully configurable between all three modes, while channels 3 and 4 are dedicated amplifier outputs.

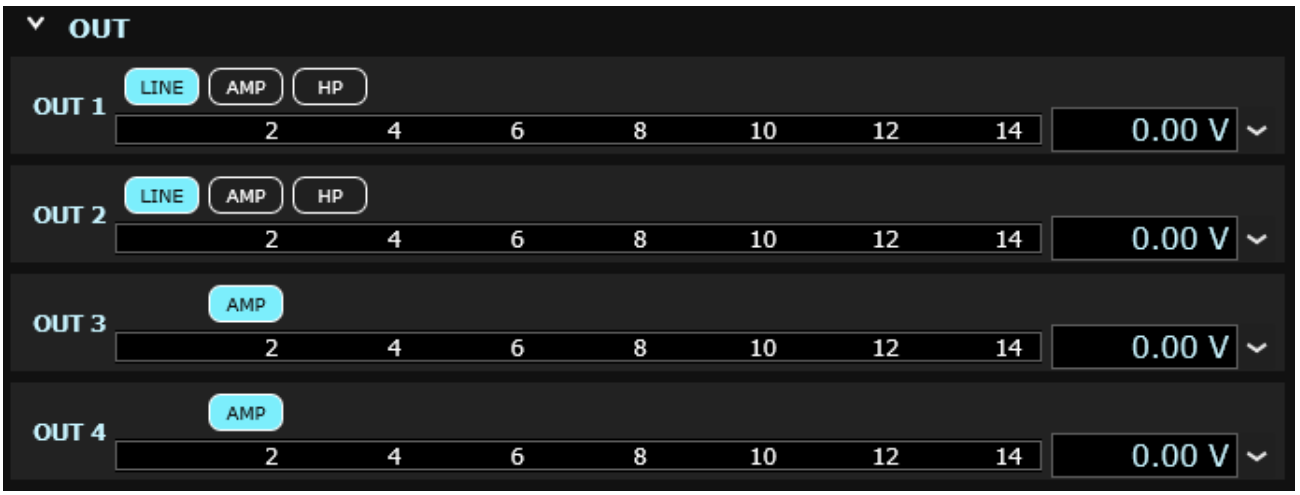

Figure 3: Channels tab - Analog Output

Output Configuration

The output level of each channel is displayed both on a horizontal bar graph and numerically. Use the drop-down control to select the desired units (dBFS, percentage, or voltage).

> ***DANGER - Mode Selection:*** *On Channels 1 and 2, click* ***LINE****,* ***AMP****, or* ***HP*** *to change the output mode. Changing the output mode adjusts the relative voltage and phase of the output port.*

Output Mode Specifications

Mode	Configuration	DC Offset	Max Output	Ground Connection
AMP	Floating differential	7.5V	12.4Vpp	Do NOT Connect
HP	Single-ended	0V	8.5Vpp	Required (shared)
LINE	Balanced	0V	22.6Vpp	Optional

Amplifier/Speaker Outputs (AMP Mode)

The AMP mode is designed for making acoustic measurements of speakers, drivers, and other acoustic transducers. This mode provides floating differential outputs with sufficient power to drive speakers directly.

Technical Specifications:	**Applications:**
• Configuration: Floating differential • DC Bias: 7.5V on both (+) and (-) terminals • Maximum Output: 27.0 Vpp • Available on: Channels 1-4 (Channels 3-4 are AMP only)	• Speaker and driver testing • Measurement microphones • Ear simulators and artificial heads • Actuators and mouth simulators

> ***WARNING:*** *Amplifier outputs are floating and not ground-referenced. Do NOT connect the ground terminal or create any low-impedance path from outputs to ground - this will cause the amplifier to shut down. The amplifier will reset automatically after the fault is removed.*

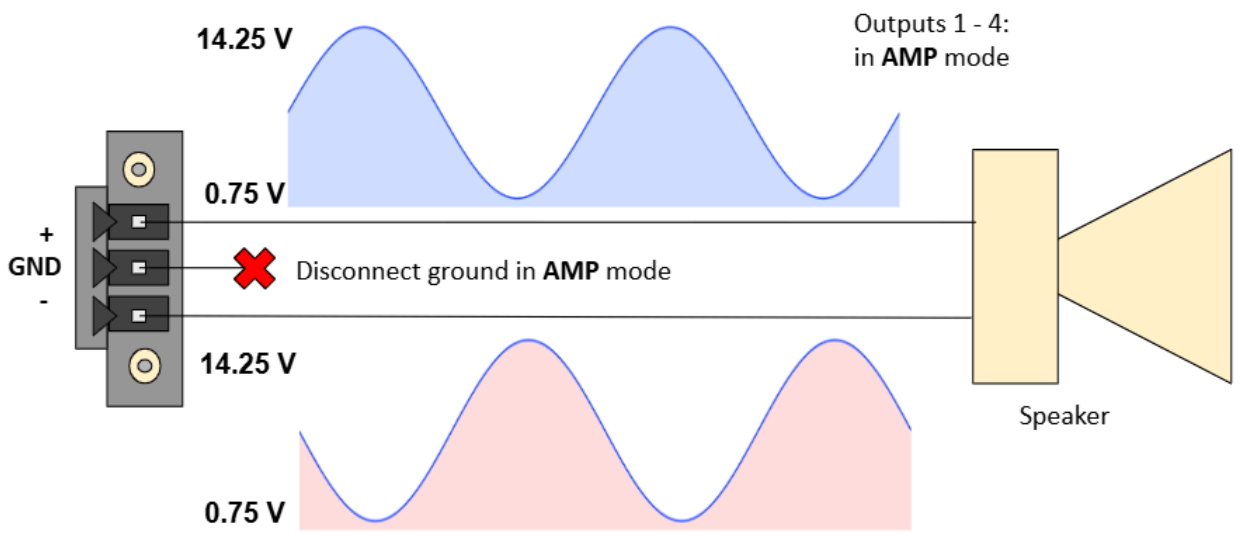

Figure 4: Connect speaker across + and - terminals only. Do NOT connect ground terminal.

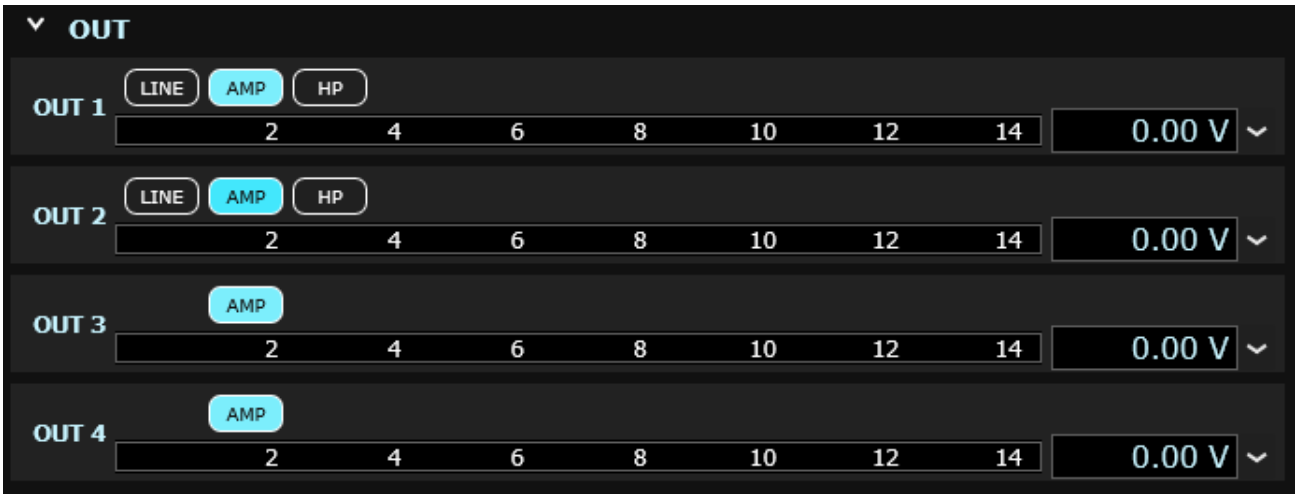

Figure 5: Control panel showing outputs 1-4 configured as AMP

Headphone Outputs (HP Mode)

The HP mode is optimized for testing wired headphones, headsets, earbuds, and hearing protection devices. This mode provides single-ended outputs with a shared ground, ideal for stereo headphone testing.

Technical Specifications:	**Applications:**
• Configuration: Single-ended • DC Offset: 0V • Maximum Output: 8.5Vpp • Ground: Required (shared between channels) • Available on: Channels 1-2 only	• Headphone and earbud testing • Headset acoustic measurements • Hearing protection device testing • Impedance measurements

> ***Note:*** *The HP mode automatically configures the outputs for proper left/right channel separation with a common ground, matching standard headphone wiring conventions.*

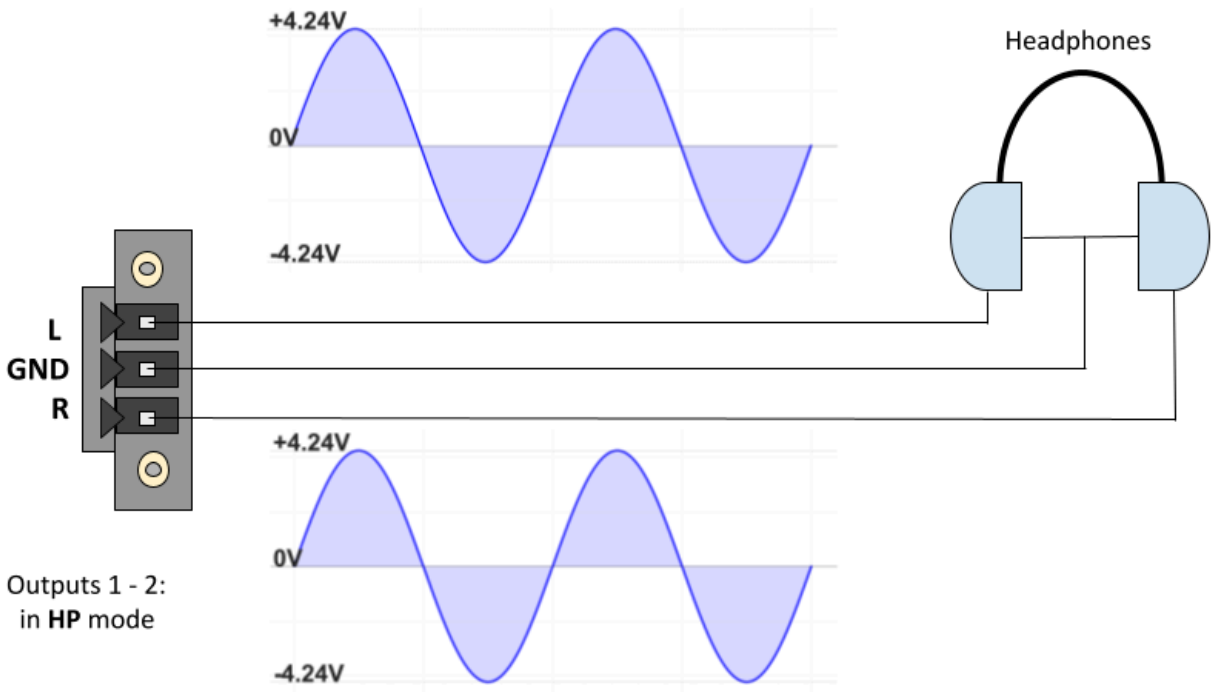

Figure 6: Headphones connected in HP mode with shared ground

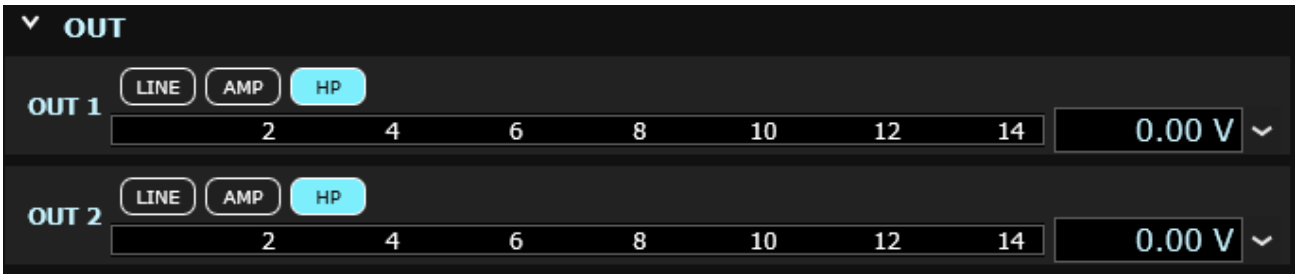

Figure 7: Control panel showing outputs 1-2 configured as HP

Line-Level Outputs (LINE Mode)

The LINE mode provides balanced outputs for interfacing with professional audio equipment and making electrical measurements. This mode offers the highest voltage output and excellent noise rejection through balanced connections.

Applications:	**Technical Specifications:**
• Professional audio equipment testing • Balanced line connections • Single-ended connections (using + and ground only) • Ground-isolated measurements	• Configuration: Balanced differential • DC Offset: 0V • Maximum Output: 22.6Vpp (balanced), 11.3Vpp (unbalanced) • Ground: Optional • Available on: Channels 1-2 only

> ***Note:*** *For single-ended (unbalanced) connections, connect to the + and ground terminals, leaving the - terminal unconnected.*

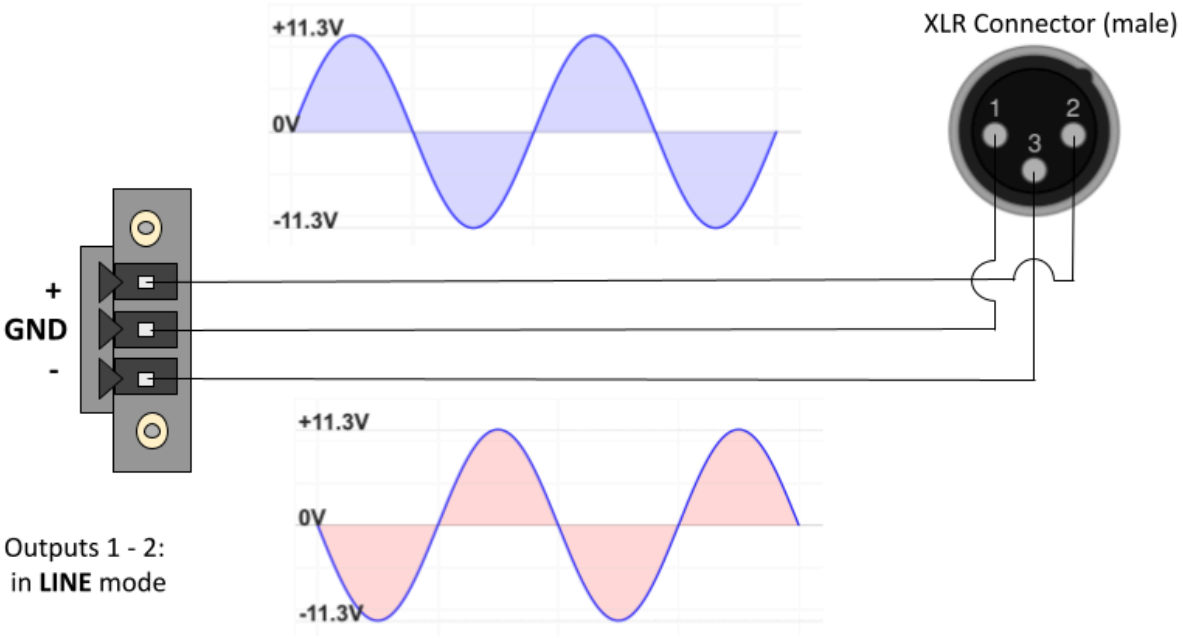

Figure 8: LINE mode connected to XLR (balanced connection shown)

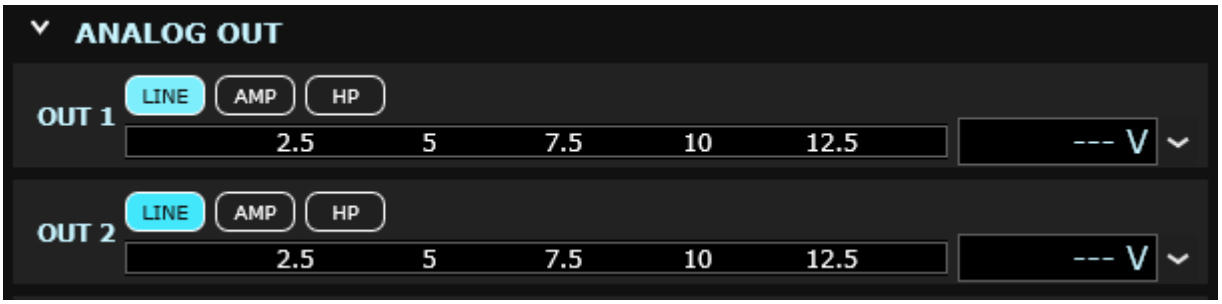

Figure 9: Control panel showing outputs 1-2 configured as LINE

5. Analog Inputs

Microphone / Line Inputs

Inputs with BNC connectors are available for connecting externally polarized or prepolarized constant current (CCP/IEPE/ICP, 4 mA) microphones, or unbalanced line-level inputs. Input gain control, constant current supply, DC coupling, and TEDS reading are available at each input and are under software control via the ATS control panel.

The mic/line inputs are designed to accommodate a very wide range of input voltages, from extremely low-level dynamic microphones all the way to line level inputs. To cover such a wide range of voltages, five input gain levels (+0 db, +10 db, +20 db, +30 db, +40 db) are available. The gain settings are individually adjustable for each channel. Keep in mind that setting the input gain higher will also amplify the noise floor, so the best practice is to use the lowest gain setting needed for a particular application. This will result in the best dynamic range for the measurements.

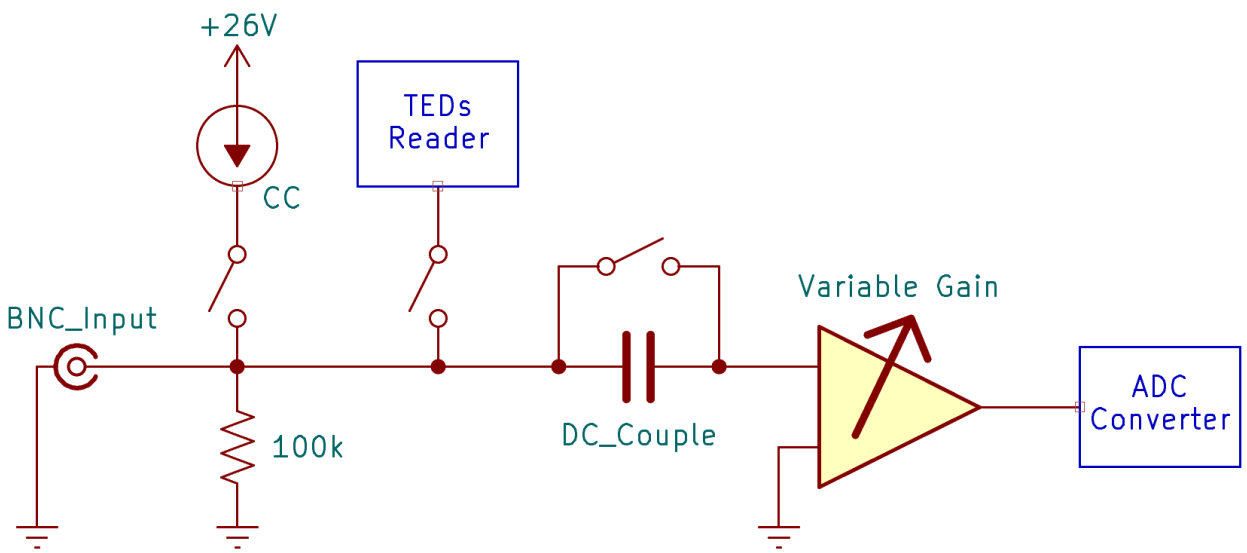

*Figure 10: Simplified Circuit Diagram of a single Input Channel**

Inputs Channel Tab

Input channels are similarly shown on the right side of the Channels tab. The input level of each channel is displayed both on a horizontal bar graph and numerically. Use the drop-down control to select the desired units.

Mic/line inputs contain the following additional controls:

Gain: Sets gain to +0 db, +10 db, +20 db, +30 db or +40 db to accommodate a wide range of signal levels.

TEDS: Displays the TEDS (Transducer Electronic Data Sheet) information reported back from the attached microphone, if the microphone supports it. See the TEDS section below for more details.

CC: Toggle the constant current supply (CCP/IEPE/ICP) to power prepolarized measurement microphones.

DC: Toggle DC coupling

Click the caret to expand or condense the channel display for each module type. In the condensed view, only the bar-graphs are visible.

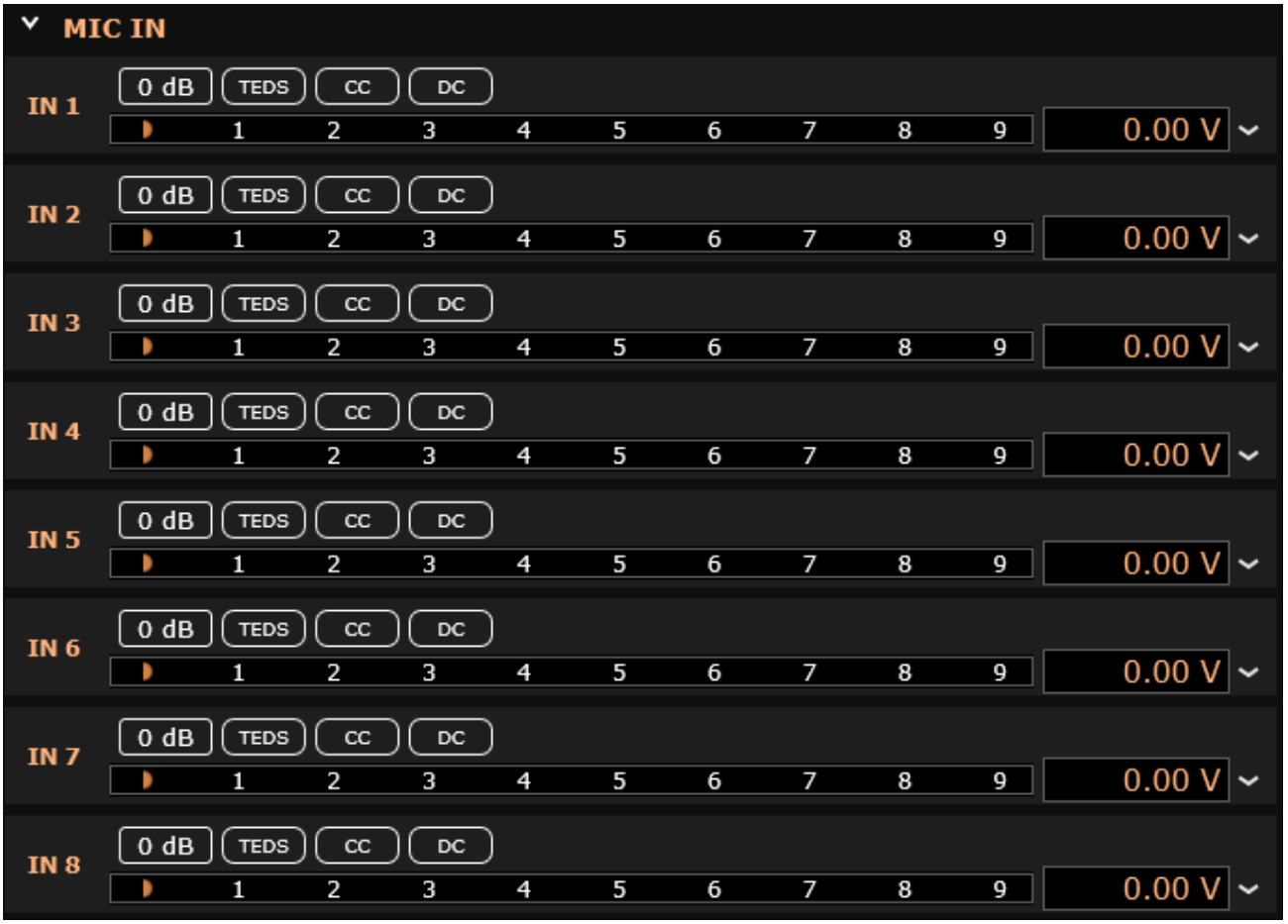

Figure 11: Channel Tab - Analog Input channels 1 - 8

6. Digital IO

The ATS includes SPDIF digital audio input and output on BNC connectors located on the rear panel. The system provides two input channels and two output channels that can operate simultaneously with the analog I/O.

Specifications

The digital interface uses consumer SPDIF format (IEC60958) with 24-bit resolution and supports standard sample rates from 44.1 kHz to 192 kHz. The SPDIF signal is self-clocking, extracting timing from the digital stream itself.

Digital channels map to the ASIO driver as:

- Input: Channels 9 and 10
- Output: Channels 5 and 6

Digital I/O Meters

In the Channels tab, the Digital In and Digital Out sections show real-time levels for channels Dig 1 and Dig 2. Levels display in either dBFS or percentage.

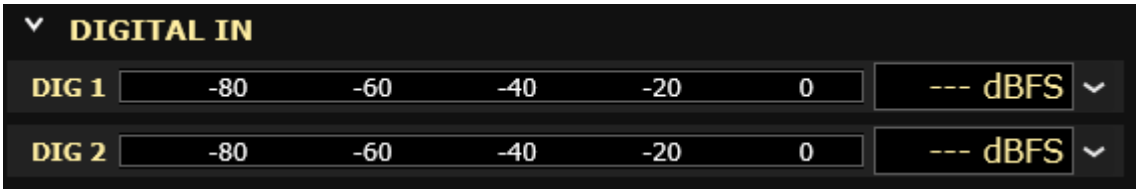

Figure 12: Channel Tab - Digital Input channel

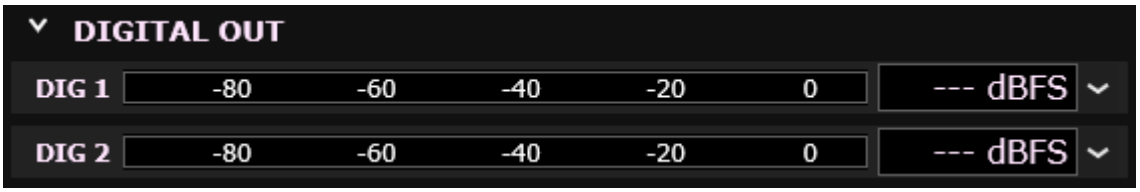

Figure 13: Channel Tab - Digital Output channel

The Options tab contains a Digital IO dropdown menu to select between:

- SPDIF Mode: Standard digital audio operation
- Word Clock Mode: Synchronization input/output

Options Tab

The Options tab contains information about the connected ATS test system and allows various configuration options to be set. Its controls and indicators are described below.

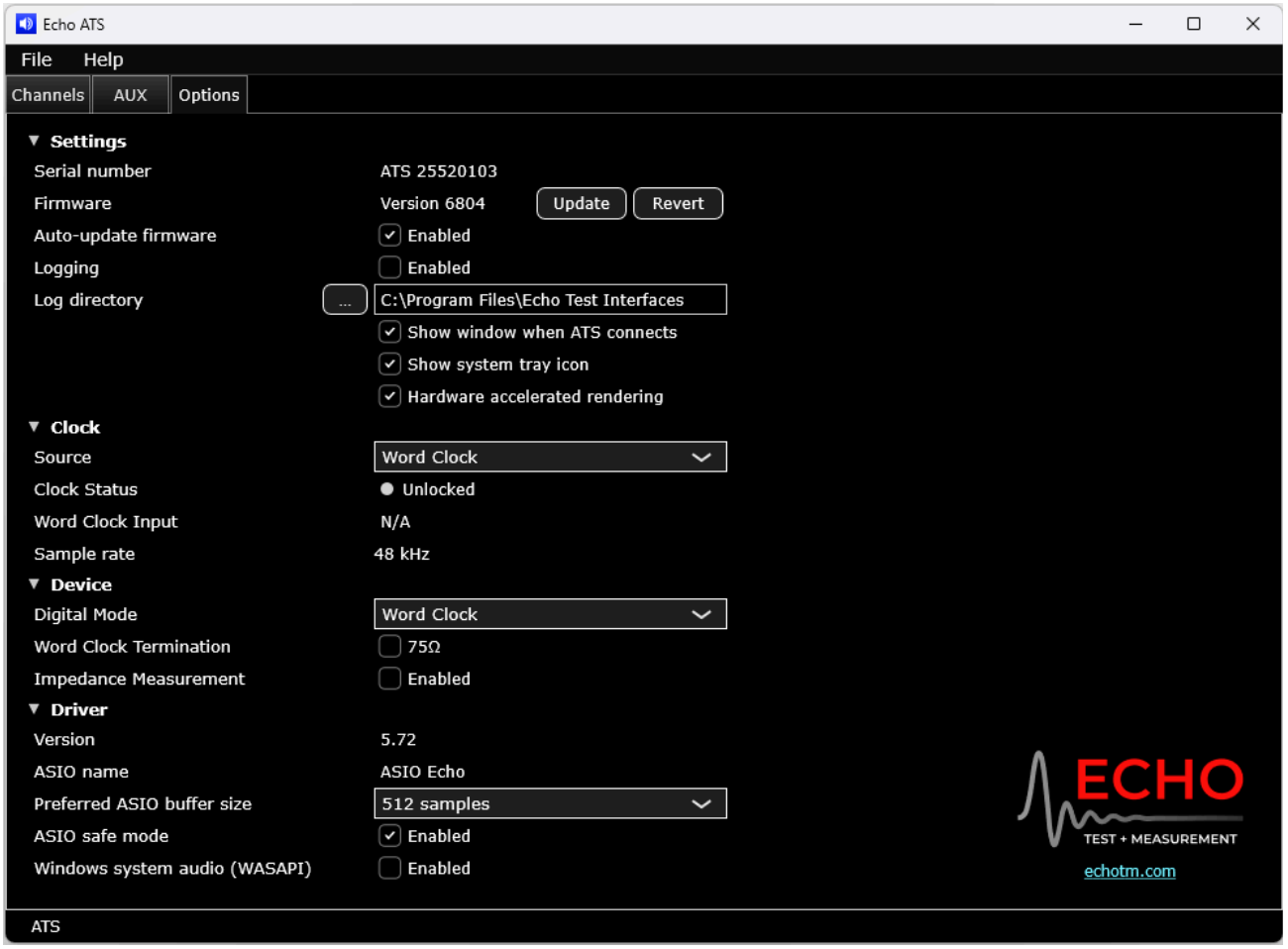

Figure 14: Options tab

Word Clock Mode

When selected from the Digital IO dropdown, the ATS configures the digital I/O BNC connectors for word clock operation. The clock can either be received from an external source (Word Clock) or generated by the ATS (Internal) for synchronizing multiple audio devices. When in Word Clock mode, the SPDIF input and output channels are disabled.

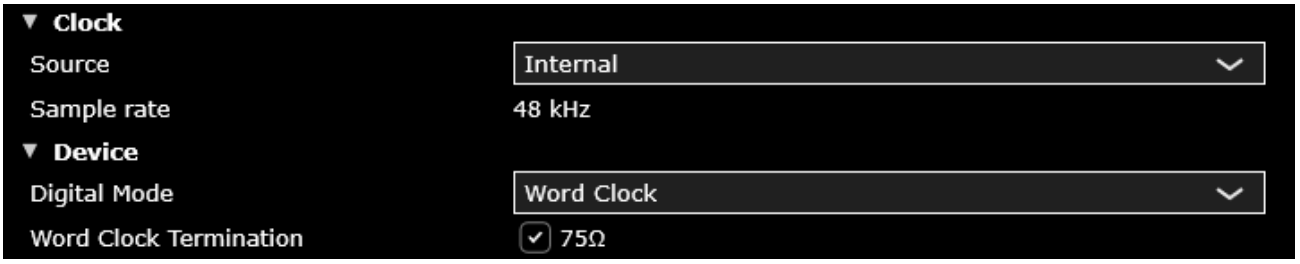

Figure 15: Options tab - Digital Mode and Clock Source

When Internal is selected, the ATS generates a word clock signal on the Digital Out BNC connector at the sample rate selected in the ASIO driver. This output can be used to synchronize other audio devices to the ATS.

When Word Clock is selected, the ATS locks to an external word clock signal applied to the Digital In BNC connector. The ATS uses this signal as its clock reference, and the sample rate is determined by the frequency of the incoming word clock.

The Options tab displays the detected input frequency when a word clock signal is present. A 75Ω termination option ensures proper impedance matching.

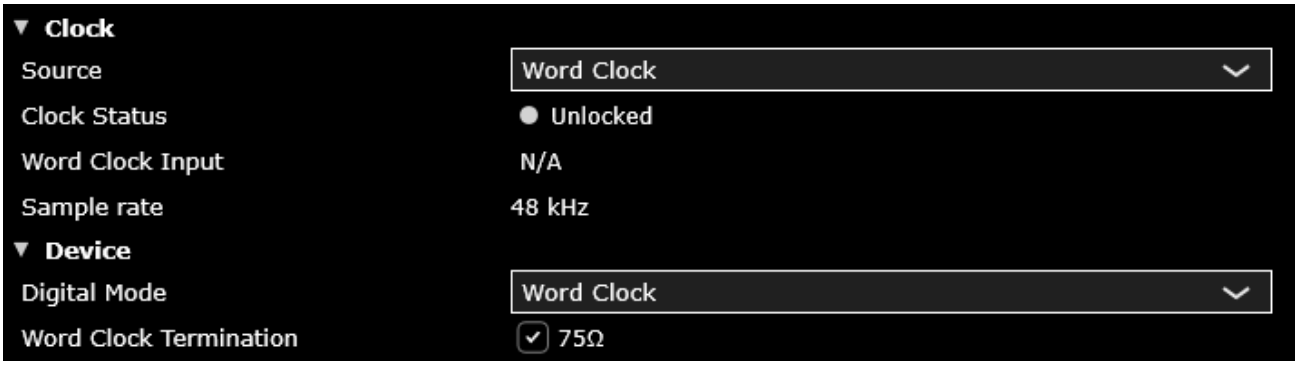

Figure 16: Options tab - Digital Mode and Clock Source - Word Clock selected

Device

- **Serial Number:** Unique serial number for the device. This can also be found on the bottom of the unit.
- **Firmware:** Displays the current firmware version and allows you to manually update or revert to the factory supplied version.
- **Auto-update firmware:** When checked, you will be prompted to update the firmware if the control panel includes a later version than is already installed.
- **Logging:** Enables diagnostic logging. You typically do not need to enable logging unless requested by Echo support.
- **Log directory:** Directory where the log file will be stored.

Clock

- **Source:** Selects the clock source as Internal (the ATS's built-in crystal oscillator).
- **Sample rate:** Shows the ASIO clock rate selected in the measurement software.

Control Panel

- **Show window when ATS connects:** Shows the control panel window when the ATS test system is connected.
- **Show system tray icon:** Shows the control panel icon in the system tray when running.
- **GPU accelerated rendering:** When checked, allows the GPU (graphics card) to offload from the CPU the task of redrawing of the control panel meters.

Driver

- **Version:** Currently installed driver version.
- **ASIO name:** Shows the name of the Echo ASIO driver as it will be displayed in the measurement software.

- **Windows system audio (WASPI):** Enables the Windows WASAPI audio driver. Leave this unchecked unless you are running software that only supports WASAPI and does not support ASIO.

File Menu

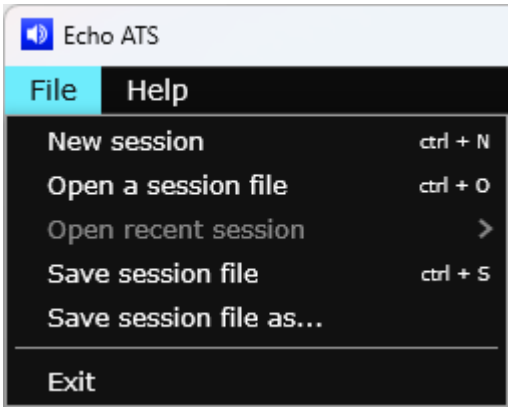

Figure 17: File menu

The File menu allows you to open and save sessions. An ATS session stores all the current hardware settings in a text file that can be used to reset the hardware to a previous state.

Additionally, the ATS command-line application can read session files, so you can fully configure the ATS, save the configuration as a session, and then later load all those settings using a single command-line call.

It is not usually necessary to directly edit the session file, as the easiest way to construct it is simply to set the control panel up as desired and then save the session. It can be, however, be easily edited in a text editor.

TEDS

When the TEDS button is clicked, the TEDS data window will be displayed. If the attached microphone does not support TEDS, then a message "TEDS device not found" will be shown instead of the data.

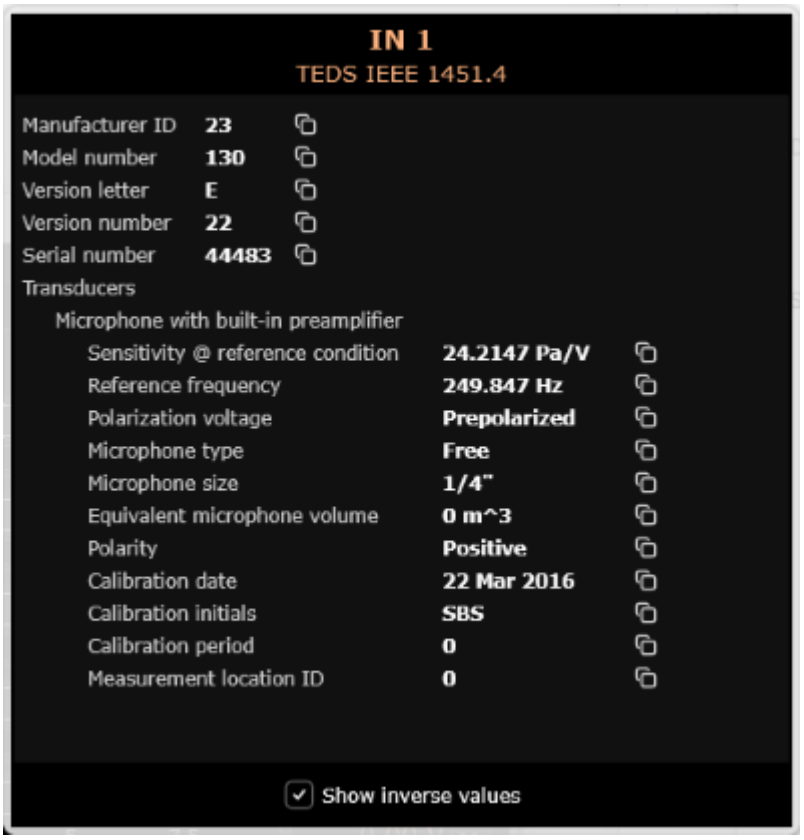

Figure 18: TEDS data window

TEDS data is not automatically transferred to the audio analysis software—it must be copied and pasted or manually entered into the appropriate field(s). Data may easily be copied by clicking the icon to the right of each line.

Sensitivity is displayed by default in units of Volt/Pascal. Some audio test software, such as Audio Precision's APx500 Flex, requires the sensitivity data to be entered in units of Pascals/Volt (also called the "scaling factor"). Checking the "Show inverse units" checkbox at the bottom will switch the "Sensitivity @ reference condition" units to Pascals/Volt.

Impedance Measurement

The ATS provides impedance measurement capability for characterizing speakers, headphones, drivers, and other loads across different frequencies. This feature works in either LINE and AMP output modes using Output 1. Impedance measurement is only available on output 1.

How It Works

The impedance measurement system uses a four-wire (Kelvin) measurement technique:

Step 1: Voltage Monitoring (VMON) Input 3 measures the voltage directly at the output terminals

Step 2: Current Monitoring (IMON) Input 4 measures current through a built-in 0.1Ω precision sense resistor

Step 3: Impedance Calculation The impedance can be calculated using Ohm's law: Z = V/I

> ***Note:*** *The current sense resistor is permanently connected in series with the positive (+) terminal of Output 1 and is not externally accessible.*

Configuration

Step 1: Connect DUT Connect your device under test to Output 1 using the three-pin connector (+, -, and ground as needed)

Step 2: Open Control Panel Navigate to the Options tab in the ATS Control Panel

Step 3: Enable Impedance Mode In the Device section, click the **Impedance Enable** checkbox to activate impedance measurement

Figure 19: Option Tab - Device in S/PDIF Digital Mode

Step 4: Verify Configuration When impedance mode is active:

- Input 3 switches to VMON (Voltage Monitor)
- Input 4 switches to IMON (Current Monitor)
- Both ADC channels are dedicated to impedance measurement in this mode

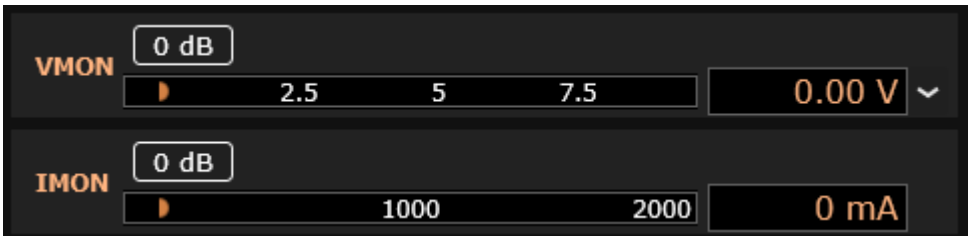

Figure 20: Channel Tab showing Inputs 1 and 2 in Impedance mode

> ***Note - Limitations:*** *Impedance measurement is available only on Output 1. When not in impedance mode, Inputs 3 and 4 function as standard Mic/Line inputs.*

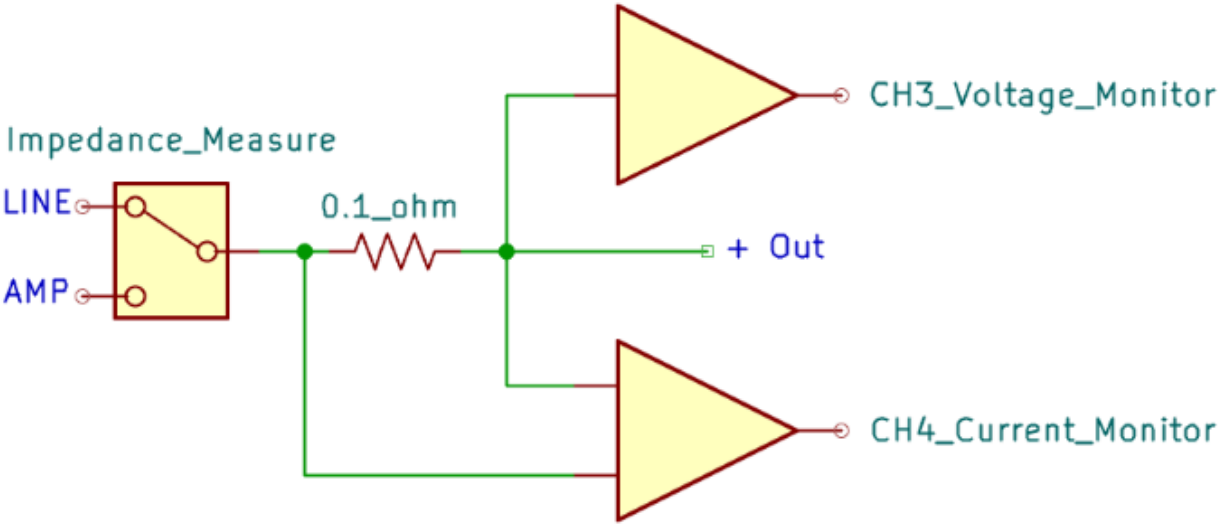

Figure 21: Simplified circuit diagram showing VMON and IMON configuration

Aux (GPIO)

The ATS features 4 TTL level inputs and 4 open drain TTL level outputs capable of sinking up to 250mA of current on a female 9 pin locking D-sub connector. They allow sending and receiving switch closure information under program control for interfacing to other test or display devices. A simple application of this feature would be to illuminate a Pass/Fail light-bar based on test results. Navigate to the Aux tab in the control panel to monitor and control the auxiliary inputs and outputs.

Figure 22: AUX Tab

DB9 Connector Pinout

Function	***Pin***
Aux Out 1	*7*
Aux Out 2	*2*
Aux Out 3	*6*
Aux Out 4	*1*
Aux In 1	*5*
Aux In 2	*9*
Aux In 3	*4*
Aux In 4	*8*
Ground	*3*

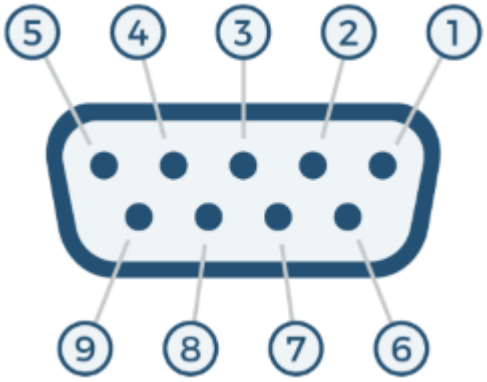

Figure 23: Female DB9 Pinout

Contact Us

Contact Echo for info or support: https://echotm.com/contact/